

AQA A-Level Further Mathematics (7367)

Question Paper

Further Calculus

E6.2 - Standard Integrals

(A-Level)

Name: _____

Class: _____

Date: _____

Time: 24 min.

Marks: 20 marks

Comments:

Q1.

- (a) Show that $\frac{1}{5 \cosh x - 3 \sinh x} = \frac{e^x}{m + e^{2x}}$, where m is an integer.

(3)

- (b) Use the substitution $u = e^x$ to show that

$$\int_0^{\ln 2} \frac{1}{5 \cosh x - 3 \sinh x} dx = \frac{\pi}{8} - \frac{1}{2} \tan^{-1} \left(\frac{1}{2} \right)$$

(5)

(Total 8 marks)

Q2.

- (a) Given that $\frac{dz}{dx} = \sinh 2x$, show that $\frac{dz}{dx} = \sinh 2x$.

(2)

- (b) Hence show that

$$\int_0^1 \frac{\sinh 2x}{1 + \cosh^4 x} dx = \tan^{-1}(\cosh^2 1) - \frac{\pi}{4}$$

(5)

(Total 7 marks)

Q3.

By using the substitution $u = x - 2$, or otherwise, find the exact value of

$$\int_{-1}^5 \frac{dx}{\sqrt{32 + 4x - x^2}}$$

(Total 5 marks)